

The drug paracetamol, also known as acetaminophen, is used to relieve pain and fever, and is a metabolite of the antipyretic drug phenacetin (Figure 1). These two drugs have similar medicinal properties, but phenacetin has been shown to be carcinogenic and cause kidney damage. Acetaminophen is a safer alternative to phenacetin when taken in therapeutic doses.

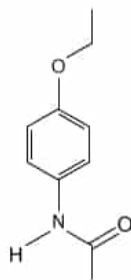
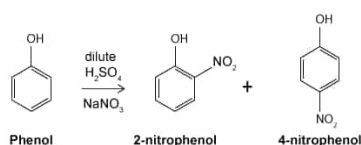
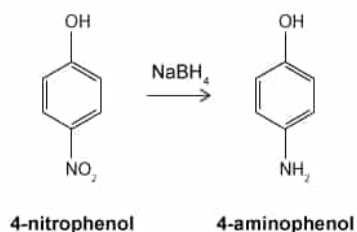


Figure 1 Structure of phenacetin

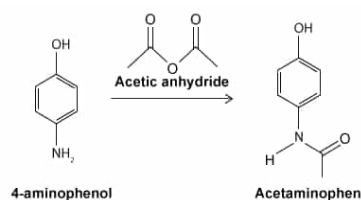
A student synthesizes acetaminophen in a laboratory from phenol according to the reactions shown. The synthesis begins with the nitration of phenol by sodium nitrate (NaNO_3) to produce the mixture of the isomers 2-nitrophenol and 4-nitrophenol (Reaction 1). These isomers can be easily separated by steam distillation to isolate the desired product, 4-nitrophenol. Reduction of the nitro group with sodium borohydride (NaBH_4) (Reaction 2) is followed by acetylation of the amine with acetic anhydride (Reaction 3) to afford the final product, acetaminophen.



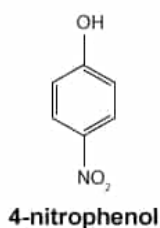
Reaction 1



Reaction 2



Reaction 3

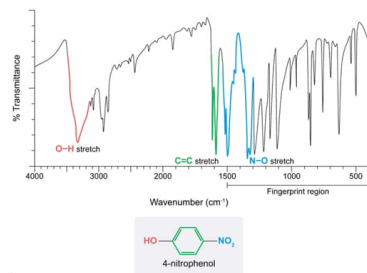


The compound 4-nitrophenol shown contains all of the following signals in its infrared spectrum EXCEPT:

- ☐ A. O–H stretch
- ☐ B. C=C stretch [3%]
- ☐ C. N–O stretch [3%]
- ☒ D. N–H stretch [92%]

Correct
92% answered correctly

Explanation:



In infrared (IR) spectroscopy, a sample is irradiated with IR light, and a spectrometer detects and records the percentage of radiation that passes through the sample over a range of frequencies. The **IR spectrum** of a compound indicates the types of bonds and **functional groups** present in a compound. Different **functional groups absorb** IR radiation at **different frequencies**, and therefore the signal from each group appears in a particular **region of the IR spectrum**. Although individual molecules may have distinct absorption spectra in the

fingerprint region (700–1500 cm⁻¹), each functional group on the molecule will still absorb in its characteristic IR region regardless of the global structural features of the molecule.

The compound 4-nitrophenol contains the following functional groups:

- Hydroxyl group (–OH)
- **Aromatic ring**
- Nitro group (–NO₂)

However, 4-nitrophenol does not contain any N–H bonds, and therefore would NOT contain an N–H stretch.

(Choice A) A hydroxyl group is present, providing an O–H stretch in the IR spectrum.

(Choice B) An aromatic ring is present, providing a C=C stretch in the IR spectrum.

(Choice C) A nitro group is present, providing an N–O stretch in the IR spectrum.

Educational objective:

An infrared (IR) spectrum displays absorption signals characteristic of the functional groups in a particular molecule based on the IR frequencies absorbed by those functional groups. Functional group bands appear in the same region of the IR spectrum regardless of the overall structure of the molecule.

Time spent: 0 sec
Subject:
Organic Chemistry

QID: 400980
System:
Separation Techniques, Spectroscopy, and Analytical Methods

The drug paracetamol, also known as acetaminophen, is used to relieve pain and fever, and is a metabolite of the antipyretic drug phenacetin (Figure 1). These two drugs have similar medicinal properties, but phenacetin has been shown to be carcinogenic and cause kidney damage. Acetaminophen is a safer alternative to phenacetin when taken in therapeutic doses.

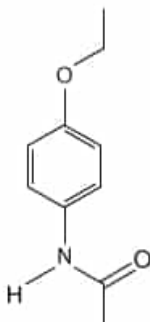
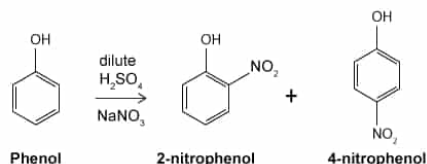
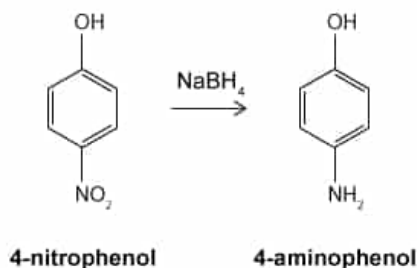


Figure 1 Structure of phenacetin

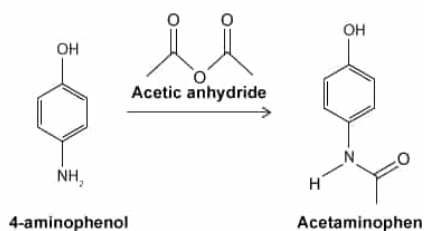
A student synthesizes acetaminophen in a laboratory from phenol according to the reactions shown. The synthesis begins with the nitration of phenol by sodium nitrate (NaNO_3) to produce the mixture of the isomers 2-nitrophenol and 4-nitrophenol (Reaction 1). These isomers can be easily separated by steam distillation to isolate the desired product, 4-nitrophenol. Reduction of the nitro group with sodium borohydride (NaBH_4) (Reaction 2) is followed by acetylation of the amine with acetic anhydride (Reaction 3) to afford the final product, acetaminophen.



Reaction 1



Reaction 2



Reaction 3

Which of the following statements most accurately describes the component that remains in the reaction flask during the steam distillation?

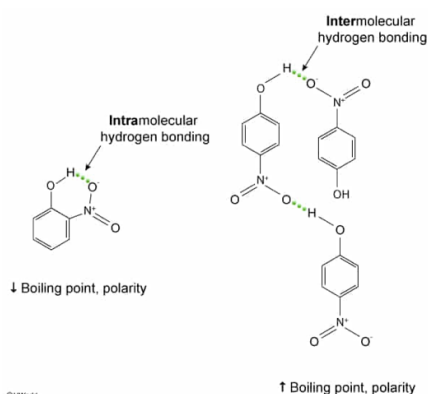
- ☐ A. 2-nitrophenol remains in the reaction flask because it has more intermolecular hydrogen bonding. [33%]
- ✓ ☒ B. 4-nitrophenol remains in the reaction flask because it has more intermolecular hydrogen bonding. [52%]

- ☐ C. 2-nitrophenol remains in the reaction flask because it has less intermolecular hydrogen bonding. [6%]
- ☐ D. 4-nitrophenol remains in the reaction flask because it has less intermolecular hydrogen bonding. [7%]

Correct

52% answered correctly

Explanation:



Distillation is a purification technique used to separate molecules based on **boiling point**. A liquid mixture is heated to a temperature that overcomes the intermolecular forces keeping the compound in the liquid phase. The vapors then condense in the collection flask as a liquid. A molecule that participates in strong intermolecular forces will have a higher boiling point than molecules with weak intermolecular forces. **Hydrogen bonding** can be *intermolecular* (between two molecules) or *intramolecular* (within the same molecule), and therefore can have a significant impact on a molecule's boiling point.

The difference in connectivity of the **constitutional isomers** 2-nitrophenol and 4-nitrophenol causes the molecules to **experience different intermolecular forces**, which contributes to the difference in their boiling points. Because the hydroxyl and nitro groups in 2-nitrophenol are **ortho** and therefore close in proximity to each other, the hydrogen from

the hydroxyl group can hydrogen bond intramolecularly with the lone pair of electrons on the nitro group. This **intramolecular** bonding decreases the number of *intermolecular* bonds that can form, thereby **decreasing the boiling point** of the compound.

Because the hydroxyl and nitro groups on 4-nitrophenol are **para**, they are able to hydrogen bond intermolecularly but not intramolecularly. **Intermolecular** bonds hold the molecules of 4-nitrophenol together, thereby **increasing the boiling point** and causing it to stay in the flask while 2-nitrophenol distills.

(Choices A and C) 2-nitrophenol experiences *less* intermolecular hydrogen bonding than 4-nitrophenol, and therefore has a *lower* boiling point and distills more rapidly than 4-nitrophenol.

(Choice D) 4-nitrophenol experiences *more* intermolecular hydrogen bonding causing it to have a *higher* boiling point than 2-nitrophenol.

Educational objective:

Distillation separates compounds based on their boiling point. Constitutional isomers can experience different intermolecular forces, contributing to the difference in their boiling points. The isomer that experiences increased *intermolecular* hydrogen bonding has a higher boiling point compared to the isomer that experiences increased *intramolecular* hydrogen bonding.

Time spent: 0 sec
Subject:
Organic Chemistry

QID: 400981
System:
Separation Techniques, Spectroscopy, and Analytical Methods

The drug paracetamol, also known as acetaminophen, is used to relieve pain and fever, and is a metabolite of the antipyretic drug phenacetin (Figure 1). These two drugs have similar medicinal properties, but phenacetin has been shown to be carcinogenic and cause kidney damage. Acetaminophen is a safer alternative to phenacetin when taken in therapeutic doses.

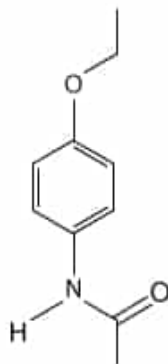
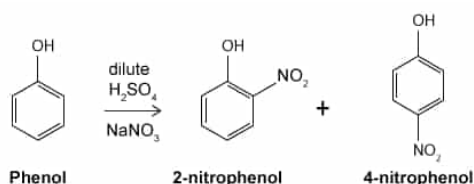
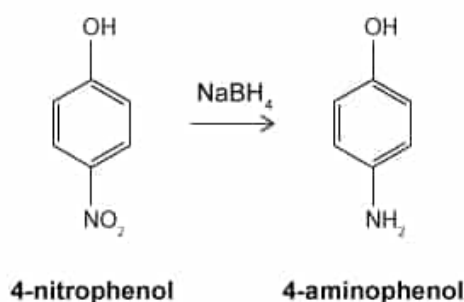


Figure 1 Structure of phenacetin

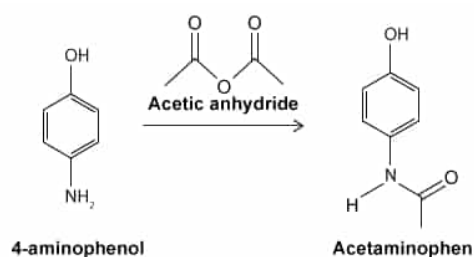
A student synthesizes acetaminophen in a laboratory from phenol according to the reactions shown. The synthesis begins with the nitration of phenol by sodium nitrate (NaNO_3) to produce the mixture of the isomers 2-nitrophenol and 4-nitrophenol (Reaction 1). These isomers can be easily separated by steam distillation to isolate the desired product, 4-nitrophenol. Reduction of the nitro group with sodium borohydride (NaBH_4) (Reaction 2) is followed by acetylation of the amine with acetic anhydride (Reaction 3) to afford the final product, acetaminophen.



Reaction 1



Reaction 2



Reaction 3

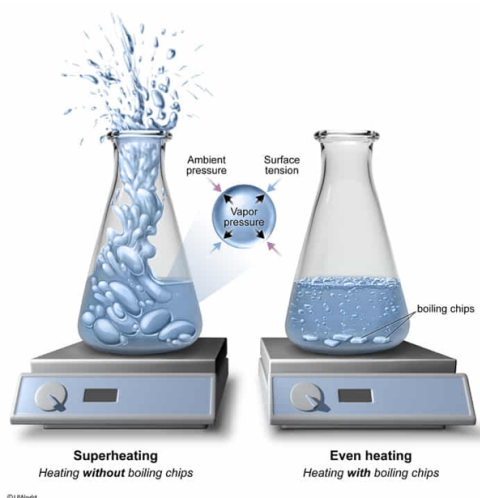
During distillation, superheating of the reaction mixture should be avoided. Which of the following would prevent superheating?

- ✓ ☒ A. Adding boiling chips to the distillation flask [62%]
- ☐ B. Increasing the rate at which the distillation flask is heated [1%]
- ☐ C. Using cold water in the condenser [12%]
- ☐ D. Heating the distillation flask under reduced pressure [23%]

Correct

62% answered correctly

Explanation:



Superheating occurs when a liquid is heated **above** its **boiling point** but does not boil. **Surface tension** can cause superheating because it can inhibit the formation of bubbles. As bubbles attempt to form, surface tension causes a local **increase** in **vapor pressure** that surpasses the ambient pressure, allowing the liquid to heat beyond its boiling point. This phenomenon can cause the formation of large bubbles at the surface, which can erupt violently and eject the hot liquid from the distillation flask in a process called bumping. This effect is difficult to overcome without scratches or crevices in the container where smaller bubbles can begin to form.

Boiling chips are made of nonreactive porous material and provide **nucleation sites** where small bubbles of vapor can form. This effect overcomes the surface tension and allows the liquid to **boil evenly** at its normal boiling temperature, thereby preventing superheating.

(Choice B) Increasing the rate at which the distillation flask is heated will cause the liquid to reach its boiling point faster but will not keep the liquid from superheating.

(Choice C) Cold water in the condenser promotes condensation of the vapors into the collection flask. This setup does not have any effect on heating the liquid to be distilled, and therefore does not prevent superheating.

(Choice D) Decreasing the pressure will decrease the boiling points of the mixture components; it has no effect on superheating.

Educational objective:

Superheating happens when a liquid is heated above its boiling point, but it does not boil. Surface tension causes the vapor pressure inside bubbles to increase as they form, causing them to explode at the surface. Addition of boiling chips gives the bubbles a surface to form on as the liquid is heated, and allows for even boiling.

Time spent:0 sec
Subject:
Organic Chemistry

QID:400982
System:
Separation Techniques, Spectroscopy, and Analytical Methods

The drug paracetamol, also known as acetaminophen, is used to relieve pain and fever, and is a metabolite of the antipyretic drug phenacetin (Figure 1). These two drugs have similar medicinal properties, but phenacetin has been shown to be carcinogenic and cause kidney damage. Acetaminophen is a safer alternative to phenacetin when taken in therapeutic doses.

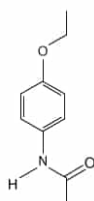
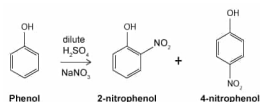
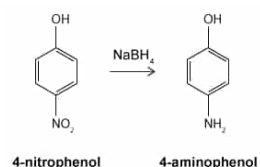


Figure 1 Structure of phenacetin

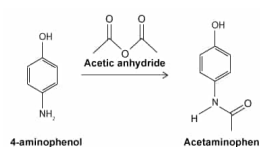
A student synthesizes acetaminophen in a laboratory from phenol according to the reactions shown. The synthesis begins with the nitration of phenol by sodium nitrate (NaNO_3) to produce the mixture of the isomers 2-nitrophenol and 4-nitrophenol (Reaction 1). These isomers can be easily separated by steam distillation to isolate the desired product, 4-nitrophenol. Reduction of the nitro group with sodium borohydride (NaBH_4) (Reaction 2) is followed by acetylation of the amine with acetic anhydride (Reaction 3) to afford the final product, acetaminophen.



Reaction 1



Reaction 2



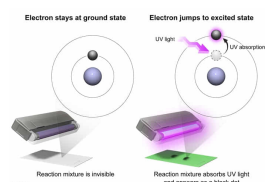
Reaction 3

Which technique could monitor the formation of 4-aminophenol by excitation of electrons with high-energy photons?

- ☐ A. Mass spectrometry [13%]
- ☐ B. ^1H NMR spectroscopy [31%]
- ☒ C. TLC with ultraviolet (UV) light [52%]
- ☐ D. Gas chromatography [2%]

Correct
52% answered correctly

Explanation:



Thin layer chromatography (TLC) is a useful technique to monitor reactions. Ultraviolet (UV) light can be used to visualize the results from TLC if the components from the reaction mixture can **absorb UV light**. UV light carries a large amount of **energy** that can **excite the electrons** of UV chromophores to a higher energy state. **UV chromophores** include double and triple bonds, carbonyls ($\text{C}=\text{O}$), **nitroso groups**, **alkyl halides**, and **conjugated systems**.

The aromatic ring in both the starting material of the reaction (4-nitrophenol) and the product (4-aminophenol) is a UV chromophore, so each compound can absorb UV light and will appear on the TLC plate.

(Choice A) In mass spectrometry, a sample is ionized by high-energy electrons, but the electrons do not change energy levels.

(Choice B) An external magnetic field is applied to a sample in NMR spectroscopy. **Radio waves**, which are low in energy, are used to detect hydrogen atoms and excite them from the α spin to the β spin state.

(Choice D) Gas chromatography requires vaporization of a sample, which is a phase change and not an electronic excitation.

Educational objective:

Absorption of ultraviolet (UV) light induces electron excitation, and a compound must contain a UV chromophore to absorb UV light.

Time spent: 0 sec
Subject:
Organic Chemistry

QID:400983
System:
Separation Techniques, Spectroscopy, and Analytical Methods

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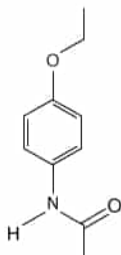
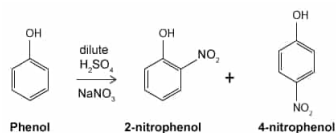
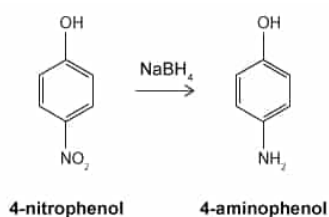


Figure 1 Structure of phenacetin

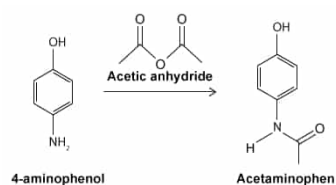
A student synthesizes acetaminophen in a laboratory from phenol according to the reactions shown. The synthesis begins with the nitration of phenol by sodium nitrate (NaNO_3) to produce the mixture of the isomers 2-nitrophenol and 4-nitrophenol (Reaction 1). These isomers can be easily separated by steam distillation to isolate the desired product, 4-nitrophenol. Reduction of the nitro group with sodium borohydride (NaBH_4) (Reaction 2) is followed by acetylation of the amine with acetic anhydride (Reaction 3) to afford the final product, acetaminophen.



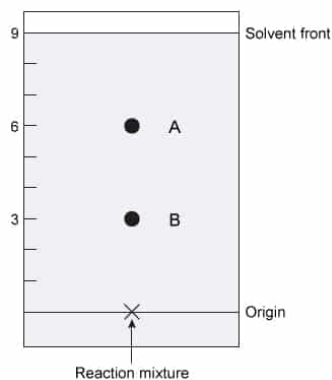
Reaction 1



Reaction 2



Reaction 3

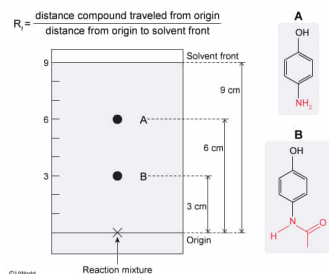


Normal phase thin layer chromatography (TLC) was used to monitor the formation of acetaminophen. According to the TLC plate shown, which spot has an R_f value of 0.33, and what compound most likely corresponds to this spot?

- ☐ A. Spot A; acetaminophen [3%]
- ☒ B. Spot B; acetaminophen [60%]
- ☐ C. Spot A; 4-aminophenol [3%]
- ☐ D. Spot B; 4-aminophenol [32%]

Correct
59% answered correctly

Explanation:



Thin layer chromatography (TLC) is a technique used to **separate** compounds based on **polarity**. The mobile phase (organic solvent) travels up the stationary phase, a thin absorbent plate coated with silica (SiO_2), via **capillary action**. The components of a mixture travel up the plate at different rates based on their **polarity**. In normal phase TLC, nonpolar compounds have less affinity for the polar stationary phase than polar compounds, and therefore travel farther up the plate.

The R_f value is expressed by the **ratio** of the distance traveled by the compound of interest to the distance traveled by the mobile phase. Values can be calculated for A and B as follows:

$$\begin{aligned} R_f &= \frac{\text{distance traveled by compound A}}{\text{distance traveled by solvent front}} \\ R_f &= \frac{\text{distance traveled by compound B}}{\text{distance traveled by solvent front}} \end{aligned}$$

An R_f value is always less than 1, and a **smaller R_f** value corresponds to a **more polar**, less mobile compound. Spot B has an R_f of 0.33 and has traveled a shorter distance up the plate than spot A, making it the more polar compound. To decide whether spot B is 4-aminophenol or acetaminophen, determine which compound is more polar.

The only difference between the two compounds is the change from amine in 4-aminophenol to **amide** in acetaminophen. An amide is a hydrogen bond donor and acceptor via two electronegative atoms (oxygen and nitrogen) whereas an **amine (primary and secondary)** is a hydrogen bond donor and acceptor via only one electronegative atom (nitrogen). Because an amide has more hydrogen bond acceptors, it is more polar; therefore, spot B must be acetaminophen.

(Choices A and C) Spot A has an R_f value of 0.67.

(Choices C and D) 4-aminophenol is the less polar compound, and therefore could not be the more polar spot B.

Educational objective:

Thin layer chromatography is used to separate compounds based on polarity. The R_f value of a compound is a ratio of the distance up the plate a compound travels to the distance the solvent travels. A polar compound will have a smaller R_f value than a nonpolar compound.

Time spent: 0 sec
Subject:
Organic Chemistry

QID: 400984
System:
Separation Techniques, Spectroscopy, and Analytical Methods

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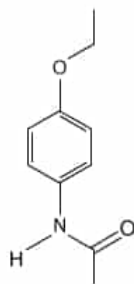
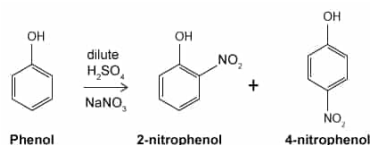
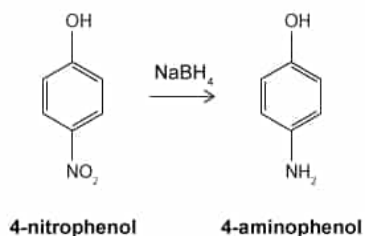


Figure 1 Structure of phenacetin

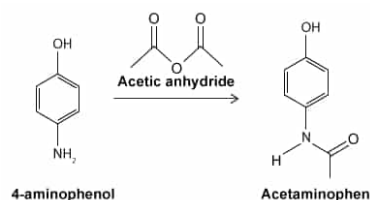
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Reaction 1



Reaction 2



Reaction 3

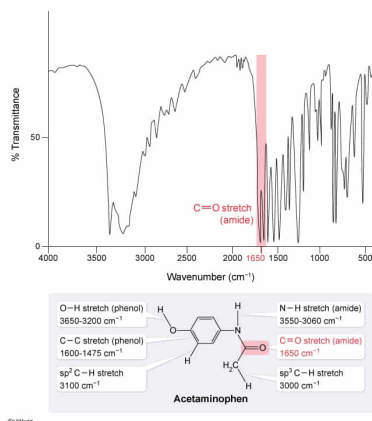
Based on the structure of acetaminophen shown in the passage, in which region of the infrared spectrum will it most likely show a strong absorption?

- ✔ ☒ A. 1690–1650 cm^{-1} [54%]
☐ B. 2150–2100 cm^{-1} [11%]
☐ C. 2260–2240 cm^{-1} [15%]
☐ D. 2850–2750 cm^{-1} [18%]

Correct

54% answered correctly

Explanation:



Acetaminophen contains several **functional groups**, including a **phenol** (hydroxyl group), **aromatic ring**, and **amide** that would show a strong **absorption** in the **infrared (IR) spectrum**. Each **functional group** shows **absorption** at a **different frequency** depending on the type of bond the functional group contains. The carbonyl (C=O) group from an **amide** shows an absorption from the 1690–1650 cm^{-1} region in the IR spectrum.

(Choices B, C, and D) In the IR spectrum, an alkyne $\text{C}\equiv\text{C}$ stretch absorbs from 2150–2,100 cm^{-1} , a nitrile $\text{C}\equiv\text{N}$ stretch absorbs from

2260–2240 cm^{-1} , and an aldehyde C–H stretch absorbs from 2850–2750 cm^{-1} . Acetaminophen does not contain these groups; therefore, it will not have a strong absorbance at these frequencies.

Educational objective:

Functional groups show absorption in the infrared spectrum at different frequencies depending on the bond type present in the particular functional group. Characteristic functional group absorptions include 3650–3200 cm^{-1} (alcohol and phenol O–H stretch); 3550–3060 cm^{-1} (amide N–H stretch); 3100 cm^{-1} (sp^2 C–H stretch); 3000–2875 cm^{-1} (sp^3 C–H stretch); 2260–2,100 cm^{-1} (triple bonds); and 1850–1650 cm^{-1} (C=O stretch).

Time spent: 0 sec

Subject:
Organic Chemistry

QID: 400985

System:
Separation Techniques, Spectroscopy, and Analytical Methods